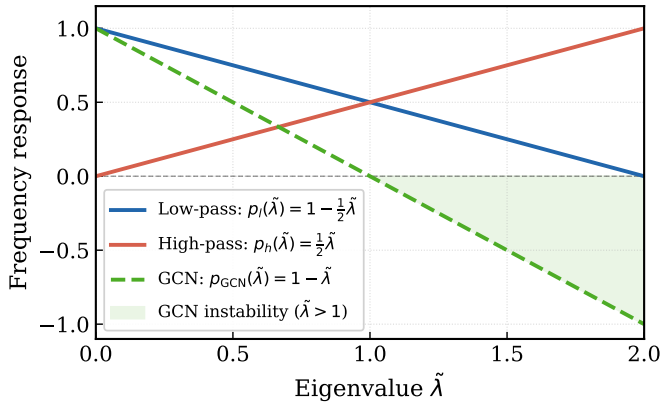
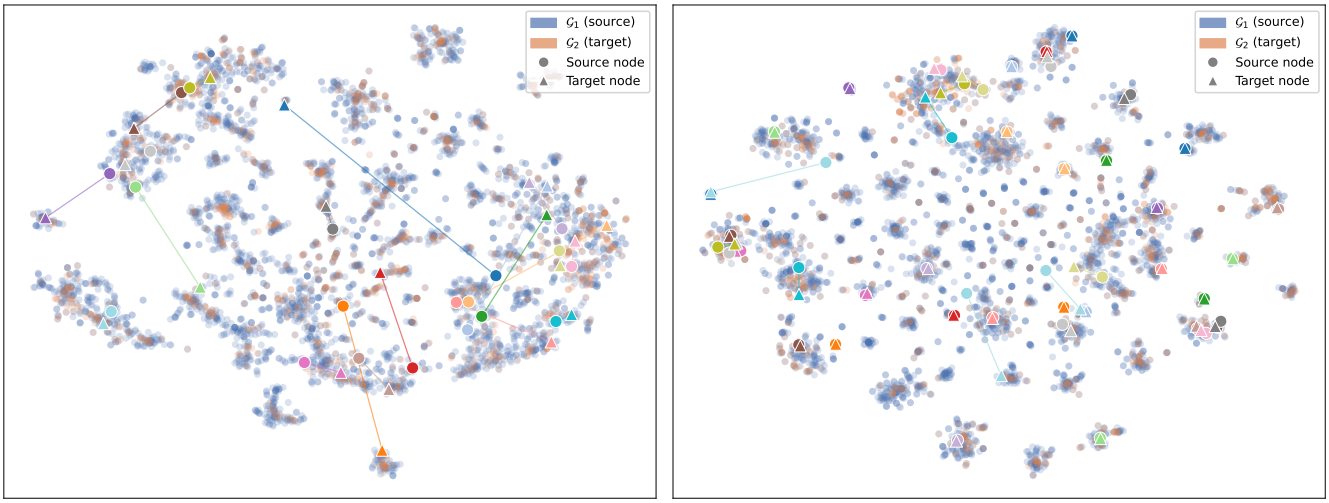


Figure 1: Spectral Frequency Response of Graph Filters



(a) Without Latent Space Alignment (Functional Map Module) *(b) With Latent Space Alignment (Functional Map Module)*

Figure 2: t-SNE visualisation of learned node embeddings (Douban Online-Offline)

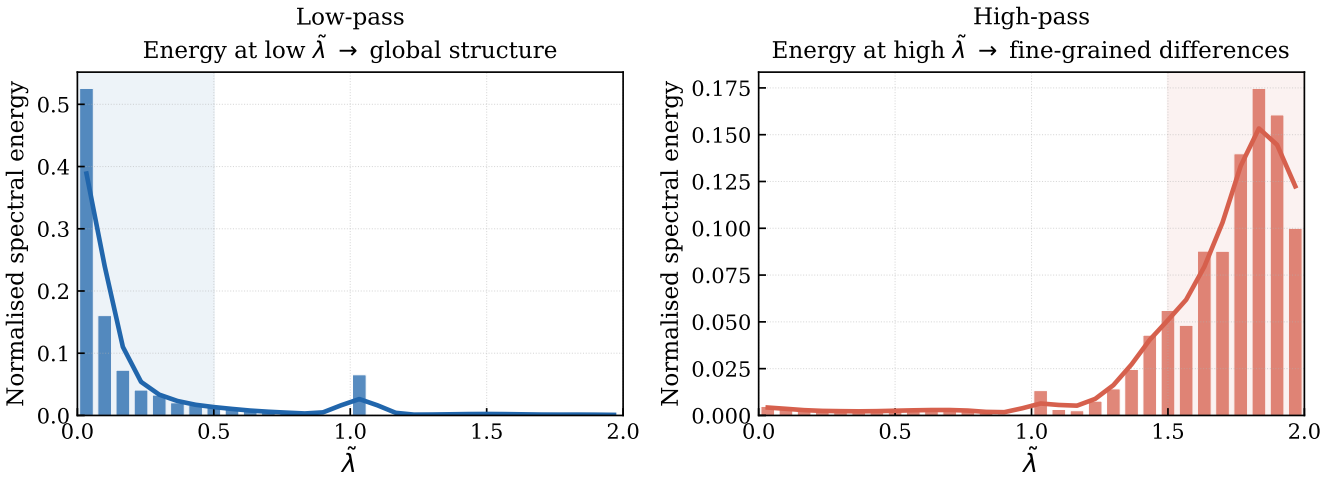
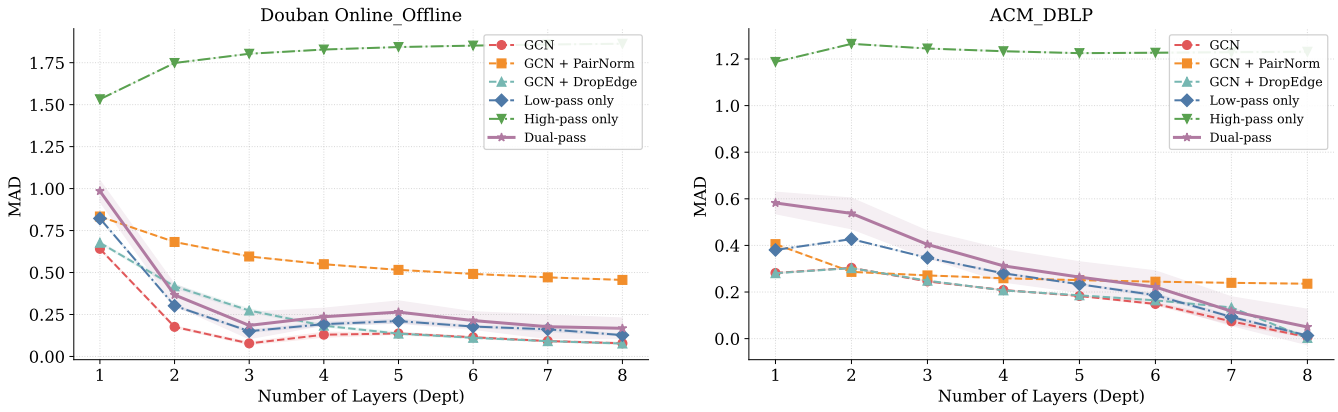


Figure 3: Spectral Energy Distribution of Node Embeddings (Douban Online-Offline)



(a) Douban Online/Offline

(b) ACM/DBLP

Figure 4: MAD vs Depth Across GNN Variants on real-world benchmarks

Table 1: Node Uniqueness Ratio Across Datasets and Encoders

Dataset	**Encoder**	$\mathbf{Z_1}$ (source)	$\mathbf{Z_2}$ (target)	**Mean**	**Rank**
Douban Online-Offline	GCN	0.0749	0.0630	0.0689	4
	Low-pass only	0.2976	0.1949	0.2463	3
	High-pass only	0.4233	0.3225	0.3729	1
	Dual-pass	0.3740	0.2599	0.3169	2
ACM-DBLP	GCN	0.1308	0.1402	0.1355	3
	Low-pass only	0.1299	0.1394	0.1346	4
	High-pass only	0.1387	0.1556	0.1471	1
	Dual-pass	0.1313	0.1407	0.1360	2
Arenas (0.05)	GCN	0.0029	0.0065	0.0047	4
	Low-pass only	0.0174	0.0169	0.0172	3
	High-pass only	0.0415	0.0395	0.0405	1
	Dual-pass	0.0255	0.0247	0.0251	2
Cora (0.05)	GCN	0.0659	0.0805	0.0732	4
	Low-pass only	0.0860	0.0866	0.0863	3
	High-pass only	0.1050	0.1063	0.1057	1
	Dual-pass	0.1028	0.1046	0.1037	2
Douban (0.05)	GCN	0.0296	0.0291	0.0293	3
	Low-pass only	0.0267	0.0263	0.0265	4
	High-pass only	0.0504	0.0581	0.0543	1
	Dual-pass	0.0349	0.0344	0.0347	2

Table 2: Performance of GADL and its variants on real-world graph alignment benchmarks

Method	**ACM-DBLP**				**Douban Online-Offline**			
	Hit@1	Hit@5	Hit@10	Hit@50	Hit@1	Hit@5	Hit@10	Hit@50
GADL w/ GCN encoder	86.34	91.24	94.13	96.68	49.28	68.79	75.51	88.91
GADL w/ low-pass only	86.17	90.16	93.38	96.34	49.47	69.81	77.52	90.29
GADL w/ high-pass only	84.17	88.28	90.30	93.38	42.58	58.59	64.85	76.57
GADL w/o bijectivity regularization	88.47	94.60	96.06	98.37	52.68	72.89	80.14	94.78
GADL w/o orthogonality regularization	88.51	94.48	96.06	98.35	51.96	72.80	79.51	94.72
GADL w/o latent-space alignment	87.42	93.50	96.03	98.34	49.35	70.23	76.72	92.66
GADL	88.63	94.76	96.16	98.41	53.31	73.61	80.67	94.18